Ex.No.13 :

Program :

```
#include <stdio.h>

// Best Fit Logic (used internally)
void bestfit(int mp[], int p[], int m, int n) {
    int j = 0;

    for (int i = 0; i < n; i++) {
        if (mp[i] >= p[j]) {
            printf("\n%d fits in %d", p[j], mp[i]);
            mp[i] = mp[i] - p[j++];
            i = -1; // restart scanning
        }
    }

    for (int i = j; i < m; i++) {
        printf("\n%d must wait for its process", p[i]);
    }
}

// Sort in descending order
void rsort(int a[], int n) {
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (a[i] > a[j]) {
                int t = a[i];
                a[i] = a[j];
                a[j] = t;
            }
        }
```

```
    }
}

// Sort in ascending order
void sort(int a[], int n) {
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (a[i] < a[j]) {
                int t = a[i];
                a[i] = a[j];
                a[j] = t;
            }
        }
    }
}

// First Fit
void firstfit(int mp[], int p[], int m, int n) {
    sort(mp, n);   // ascending
    sort(p, m);
    bestfit(mp, p, m, n);
}

// Worst Fit
void worstfit(int mp[], int p[], int m, int n) {
    rsort(mp, n);  // descending
    sort(p, m);
    bestfit(mp, p, m, n);
}

int main() {
```

```
int m, n, mp[20], p[20], ch;

printf("Number of memory partitions: ");
scanf("%d", &n);

printf("Number of processes: ");
scanf("%d", &m);

printf("Enter memory partitions:\n");
for (int i = 0; i < n; i++) {
   scanf("%d", &mp[i]);
}

printf("Enter process sizes:\n");
for (int i = 0; i < m; i++) {
   scanf("%d", &p[i]);
}

printf("\n1. First Fit\t2. Best Fit\t3. Worst Fit\n");
printf("Enter your choice: ");
scanf("%d", &ch);

switch (ch) {
   case 1:
      firstfit(mp, p, m, n);
      break;

   case 2:
      bestfit(mp, p, m, n);
      break;
```

```
        case 3:
            worstfit(mp, p, m, n);
            break;

        default:
            printf("Invalid choice\n");
    }

    return 0;
}
```

Output

```
Number of memory partition : 5
Number of process : 4
Enter the memory partitions :
150
220
500
350
700
ENter process size :
160
450
500
412
1. Firstfit 2. Bestfit  3. worstfit
Enter your choice : 1

160 fits in 220
450 fits in 500
500 fits in 700
412 must wait for its process

=== Code Execution Successful ===
```